test-cube-root_3

October 12, 2026

[12]:
```
prec = 3400
CC = ComplexBallField(prec)
RBF = RealBallField(prec)
Pi = CC(pi, 0)
load ('cubic_field_K_cube_root_3.sage')
load('BCG_cubic_git.sage')
```

```
Loaded 'cubic_field_K_cube_root_3.sage'. Contains the following variables:

K: Number Field in y with defining polynomial y^3 - 3
K has discriminant  -243 = -1 * 3^5
K has class number  1

pol_K = defining polynomial =  y^3 - 3
K_basis_SAGE = basis of K =  [1, y, y^2]

yRR = unique real root of pol_K  =  1.44224957030741 …
yCC = unique complex root of pol_K with positive imaginary part
    =  -0.721124785153704 … + I *  1.24902476648341 …
sigmaC = complex embedding of K sending y to yCC
K_basis_SAGE_sigmaC = [1, yCC, yCC^2]
K_basis_SAGE_sigmaR = [1, yRR, yRR^2]

ff =  Fractional ideal (y)  of norm  3
q_ff = smallest positive integer in ff = 3
The narrow ray class group of conductor ff has order 2.

aa =  Fractional ideal (y - 2)
N_aa = norm of aa =  5
aa_comp = N_aa * aa^(-1)
bb =  Fractional ideal (y - 1)  of norm  2
(bb generates the narrow ray class group modulo ff.)

vareps = generator of totally positive units in 1+ff =  y^2 - 2
varepsRR = real image of vareps =  0.0800838230519041
varepsCC = image of vareps under sigmaC
or_matrix = Matrix with entries the real and complex images of K_basis_SAGE
Eps =  matrix of the fundamental unit vareps in the basis K_basis_SAGE =
```

```
[-2  0  1]
[ 3 -2  0]
[ 0  3 -2]

N(aa) = 5 is an odd prime, as (2.9) asks.
aa is coprime to ff, as (2.9) asks.
sigmaR(vareps) = 0.080084 lies in (0,1), as section 2.2 asks.
vareps lies in 1 + ff, as section 2.2 asks.
q_ff = 3 is a positive integer of ff, as section 2.2 asks.
```

[13]:
```
%%time
L = ff
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho         N_Ell_Gamma  adm
49.645     3     0.060429    38105        (3, 24, -18)
49.645     3     0.060429    38105        (-6, -3, 6)
49.645     3     0.060429    38105        (12, 6, 3)
99.290     3     0.030214    76209        (3, 9, -3)
99.290     3     0.030214    76209        (21, -27, 9)
CPU times: user 1.45 s, sys: 32 ms, total: 1.48 s
Wall time: 1.48 s
```

[16]:
```
%%time
N_Ell_Gamma = 39000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 6, certified_digits(u) - 5)
PK=algdep_K(u, 2, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (3, 24, -18)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (1, 8, -6)  with adm = (3, 24, -18)
The wedge for  (1, 8, -6)  is
a =  (-2, -2, -3)
b =  (-20, -29, -42)
alpha = (0, 42, -29) , alpha(b) =  0 alpha(a) =  3
beta = (0, -3, 2) , beta(a) =  0 beta(b) =  3
gamma = (-3, -24, 18)
(alpha,beta,gamma) is positively oriented basis
tau =  -1.6693361274 - 0.12009369552*I
```

```
sigma =  0.11710278810 + 0.0096175622615*I
The set F has  3 elements. They are
[  0  26 -18]
[  0  13  -9]
[  0   0   0]
(z, tau, sigma) in K =  [2/15*y^2 + 2/15*y - 17/15, 1/15*y^2 - 8/5, 2/15*y^2 +
1/5*y + 2/5]
(z, tau, sigma) in K =  [1/15*y^2 + 1/15*y - 11/15, 1/15*y^2 - 8/5, 2/15*y^2 +
1/5*y + 2/5]
(z, tau, sigma) in K =  [-1/3, 1/15*y^2 - 8/5, 2/15*y^2 + 1/5*y + 2/5]
Wedge negatively oriented with respect to h
certified decimals =  1016
u= 0.18108287362775213 + 0.21746911488903353*I
1/u= 2.2611666966796562 - 2.7155186478558465*I
P= x^6 - 3*x^5 + 6*x^4 + 17*x^3 + 6*x^2 - 3*x + 1

PK= x^2 + (2*y - 1)*x + 1
CPU times: user 47.9 s, sys: 0 ns, total: 47.9 s
Wall time: 47.9 s
```

[17]:
```
%%time
L = ff*bb^(-1)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho       s      rho          N_Ell_Gamma  adm
49.645      3      0.060429     38105        (-3/2, -9/2, 3/2)
49.645      3      0.060429     38105        (-21/2, 27/2, -9/2)
49.645      3      0.060429     38105        (-39/2, -27/2, -21/2)
99.290      3      0.030214     76209        (-21/2, -3/2, -9/2)
99.290      3      0.030214     76209        (33/2, -21/2, -3/2)
CPU times: user 1.55 s, sys: 35.9 ms, total: 1.59 s
Wall time: 1.59 s
```

[18]:
```
%%time
N_Ell_Gamma = 39000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u^4, 6, certified_digits(u) - 5)
PK=algdep_K(u^4, 2, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (-3/2, -9/2, 3/2)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
```

```
Evaluating at h = adm/3 = (-1/2, -3/2, 1/2)  with adm = (-3/2, -9/2, 3/2)
The wedge for  (-1/2, -3/2, 1/2)  is
a =  (-3, 2, 3)
b =  (0, -1, -3)
alpha = (-3/2, -3/2, 1/2) , alpha(b) =  0 alpha(a) =  3
beta = (-3, -6, 1) , beta(a) =  0 beta(b) =  3
gamma = (-3/2, -9/2, 3/2)
(alpha,beta,gamma) is positively oriented basis
tau =  0.33066387256 - 0.12009369552*I
sigma =  1.1171027881 + 0.0096175622615*I
The set F has  3 elements. They are
[  -3   -5    1]
[-3/2 -5/2  1/2]
[   0    0    0]
(z, tau, sigma) in K =  [2/15*y^2 + 2/15*y + 23/15, 1/15*y^2 + 2/5, 2/15*y^2 +
1/5*y + 7/5]
(z, tau, sigma) in K =  [1/15*y^2 + 1/15*y + 14/15, 1/15*y^2 + 2/5, 2/15*y^2 +
1/5*y + 7/5]
(z, tau, sigma) in K =  [1/3, 1/15*y^2 + 2/5, 2/15*y^2 + 1/5*y + 7/5]
Wedge positively oriented with respect to h
certified decimals =  1016
u= 2.2611666966796562 - 2.7155186478558465*I
1/u= 0.18108287362775213 + 0.21746911488903353*I
P= x^6 + 291*x^5 + 24198*x^4 - 9305*x^3 + 24198*x^2 + 291*x + 1

PK= x^2 + (-8*y^2 - 56*y + 97)*x + 1
CPU times: user 48 s, sys: 0 ns, total: 48 s
Wall time: 48 s
```

[19]:
```
# 2nd run, now with the smoothing aa of norm 2 (change in master file)
  ↪cubic_field_cube_root_3
prec = 3400
CC = ComplexBallField(prec)
RBF = RealBallField(prec)
Pi = CC(pi, 0)
load ('cubic_field_K_cube_root_3.sage')
load('BCG_cubic_git.sage')
```

```
Loaded 'cubic_field_K_cube_root_3.sage'. Contains the following variables:

K: Number Field in y with defining polynomial y^3 - 3
K has discriminant  -243 = -1 * 3^5
K has class number  1

pol_K = defining polynomial =  y^3 - 3
K_basis_SAGE = basis of K =  [1, y, y^2]

yRR = unique real root of pol_K  =  1.44224957030741 …
```

```
yCC = unique complex root of pol_K with positive imaginary part
    =  -0.721124785153704 … + I *  1.24902476648341 …
sigmaC = complex embedding of K sending y to yCC
K_basis_SAGE_sigmaC = [1, yCC, yCC^2]
K_basis_SAGE_sigmaR = [1, yRR, yRR^2]

ff =  Fractional ideal (y)  of norm  3
q_ff = smallest positive integer in ff = 3
The narrow ray class group of conductor ff has order 2.

aa =  Fractional ideal (y - 1)
N_aa = norm of aa =  2
aa_comp = N_aa * aa^(-1)
bb =  Fractional ideal (y - 1)  of norm  2
(bb generates the narrow ray class group modulo ff.)

vareps = generator of totally positive units in 1+ff =  y^2 - 2
varepsRR = real image of vareps =  0.0800838230519041
varepsCC = image of vareps under sigmaC
or_matrix = Matrix with entries the real and complex images of K_basis_SAGE
Eps =  matrix of the fundamental unit vareps in the basis K_basis_SAGE =
[-2  0  1]
[ 3 -2  0]
[ 0  3 -2]

PROBLEM: N(aa) = 2 should be an odd prime, see (2.9).
aa is coprime to ff, as (2.9) asks.
sigmaR(vareps) = 0.080084 lies in (0,1), as section 2.2 asks.
vareps lies in 1 + ff, as section 2.2 asks.
q_ff = 3 is a positive integer of ff, as section 2.2 asks.
```

[20]:
```
%%time
L = ff
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho         N_Ell_Gamma  adm
19.858     3     0.15107     15242        (3, 3, -3)
19.858     3     0.15107     15242        (3, -15, 9)
19.858     3     0.15107     15242        (3, 3, 3)
39.716     3     0.075536    30484        (3, -3, 3)
39.716     3     0.075536    30484        (-15, 15, -3)
CPU times: user 7.14 s, sys: 112 ms, total: 7.25 s
Wall time: 7.25 s
```

[23]:
```
%%time
N_Ell_Gamma = 16000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u^4, 6, certified_digits(u) - 5)
PK=algdep_K(u^4, 2, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
print(L == ff, abs(u))
```

```
adm= (3, 3, -3)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (1, 1, -1)  with adm = (3, 3, -3)
The wedge for  (1, 1, -1)  is
a =  (-1, 1, 0)
b =  (-1, -2, -3)
alpha = (0, 3, -2) , alpha(b) =  0 alpha(a) =  3
beta = (0, 0, -1) , beta(a) =  0 beta(b) =  3
gamma = (-3, -3, 3)
(alpha,beta,gamma) is positively oriented basis
tau =  -0.67334031859 - 0.30023423879*I
sigma =  -0.20724302975 + 0.024043905654*I
The set F has  3 elements. They are
[ 0  2 -2]
[ 0  1 -1]
[ 0  0  0]
(z, tau, sigma) in K =  [1/3*y^2 + 1/3*y - 1/3, 1/6*y^2 - 1/2, 1/3*y^2 + 1/2*y +
1/2]
(z, tau, sigma) in K =  [1/6*y^2 + 1/6*y - 1/3, 1/6*y^2 - 1/2, 1/3*y^2 + 1/2*y +
1/2]
(z, tau, sigma) in K =  [-1/3, 1/6*y^2 - 1/2, 1/3*y^2 + 1/2*y + 1/2]
Wedge negatively oriented with respect to h
certified decimals =  1018
u= 0.34271200883057569 - 0.40686525079452576*I
1/u= 1.2110356389021495 + 1.4377328668007624*I
P= x^6 + 24*x^5 + 168*x^4 + 98*x^3 + 168*x^2 + 24*x + 1

PK= x^2 + (-3*y^2 - y + 8)*x + 1
True [0.531968846175018911446522925892756154558680399553858624707739117729944480
585150293744820655203284284301304508773617506058482168478668672839075435812038811
702669795417167036690547277781047361618528259889655658616968724343951707563964 7
970040811531704350680300582111036989180538227448548553500882432286918448473061749
495115109638019822124790164623068971056662412920357139616164133690431589320141417
091660992678837646402192030864571728966466650338717464606264447487234333043076828
869554377599621662035839673751622299279398757621529801294069497730349656781642 0
```

```
607114158895027772710575377052449788169033647437384891787154462567828960278335912340034907119665621947060616472739215258879921383598656685461717304306088525976272277114235767306328490073162114223579035874675841673707227998607947519785125827205730881982316005899262361429904264706289355073282613706536312489556366882129465723711759831730919286241571181984507226643740057184208128666181373726252221776760517412429776357764519170353974398271663288979896139612499663368 7 +/-
4.73e-1019]
CPU times: user 19.9 s, sys: 167  s, total: 19.9 s
Wall time: 19.9 s
```

[24]:
```
%%time
L = ff*bb^(-1)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho         N_Ell_Gamma  adm
19.858     3     0.15107     15242        (-3/2, 3/2, -3/2)
19.858     3     0.15107     15242        (15/2, -15/2, 3/2)
19.858     3     0.15107     15242        (-21/2, -15/2, -9/2)
19.858     3     0.15107     15242        (-75/2, 39/2, 9/2)
39.716     3     0.075536    30484        (-3/2, -9/2, -3/2)
CPU times: user 7.01 s, sys: 96 ms, total: 7.1 s
Wall time: 7.1 s
```

[26]:
```
%%time
N_Ell_Gamma = 16000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u^4, 6, certified_digits(u) - 5)
PK=algdep_K(u^4, 2, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (-3/2, 3/2, -3/2)
Did not have to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (-1/2, 1/2, -1/2)  with adm = (-3/2, 3/2, -3/2)
The wedge for  (-1/2, 1/2, -1/2)  is
a =  (2, -1, -3)
b =  (-1, -1, 0)
alpha = (3/2, -3/2, 1/2) , alpha(b) =  0 alpha(a) =  3
beta = (-3/2, -3/2, -1/2) , beta(a) =  0 beta(b) =  3
gamma = (-3/2, 3/2, -3/2)
(alpha,beta,gamma) is positively oriented basis
tau =  -0.67334031859 - 0.30023423879*I
sigma =  -0.20724302975 + 0.024043905654*I
```

```
The set F has  3 elements. They are
[ 0 -2  0]
[ 0 -1  0]
[ 0  0  0]
(z, tau, sigma) in K =  [1/3*y^2 + 1/3*y + 1/3, 1/6*y^2 - 1/2, 1/3*y^2 + 1/2*y +
1/2]
(z, tau, sigma) in K =  [1/6*y^2 + 1/6*y + 1/3, 1/6*y^2 - 1/2, 1/3*y^2 + 1/2*y +
1/2]
(z, tau, sigma) in K =  [1/3, 1/6*y^2 - 1/2, 1/3*y^2 + 1/2*y + 1/2]
Wedge positively oriented with respect to h
certified decimals =  1017
u= 1.4377328668007624 - 1.2110356389021495*I
1/u= 0.40686525079452576 + 0.34271200883057569*I
P= x^6 + 24*x^5 + 168*x^4 + 98*x^3 + 168*x^2 + 24*x + 1

PK= x^2 + (-3*y^2 - y + 8)*x + 1
CPU times: user 19.7 s, sys: 3.98 ms, total: 19.7 s
Wall time: 19.7 s
```

[ ]: